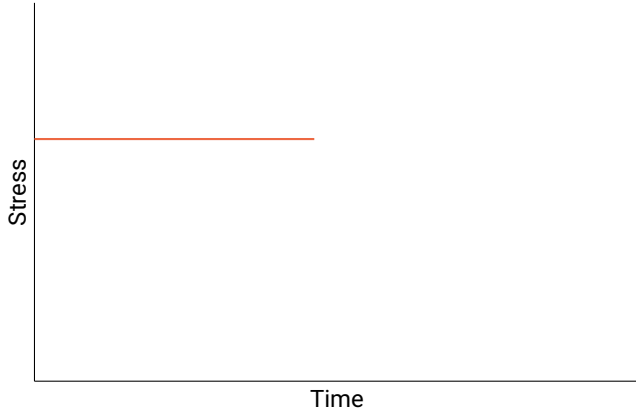


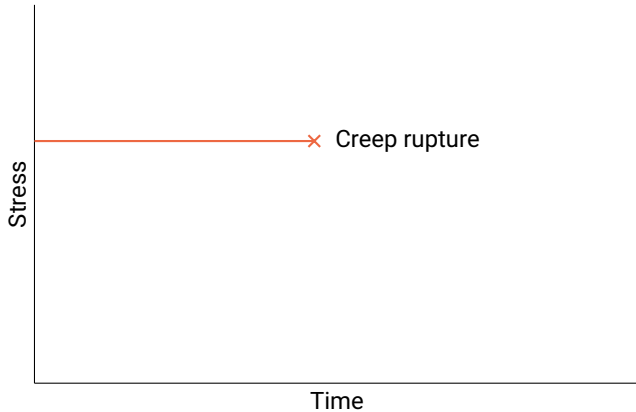
Microscale modeling of creep and fatigue in composites: viscoplasticity, cohesive damage and time homogenization

Dragan Kovačević, Pieter Hofman, Juri Rocha, Frans van der Meer

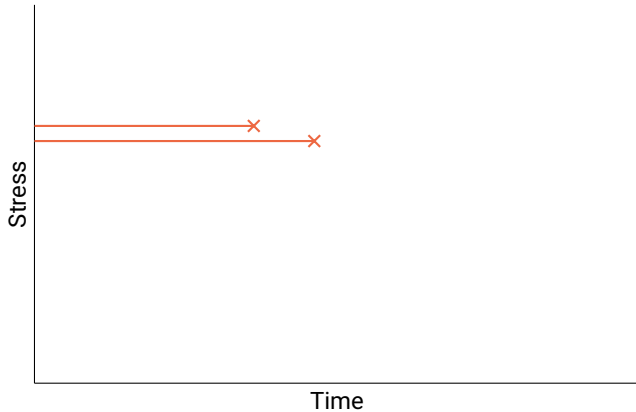
Cyclic loading has a two-fold effect on the lifetime of polymers



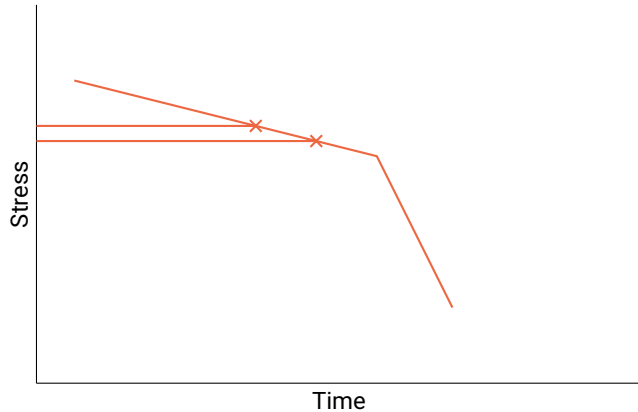
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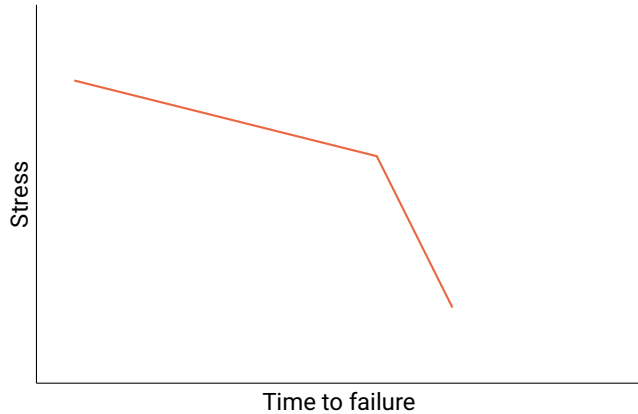
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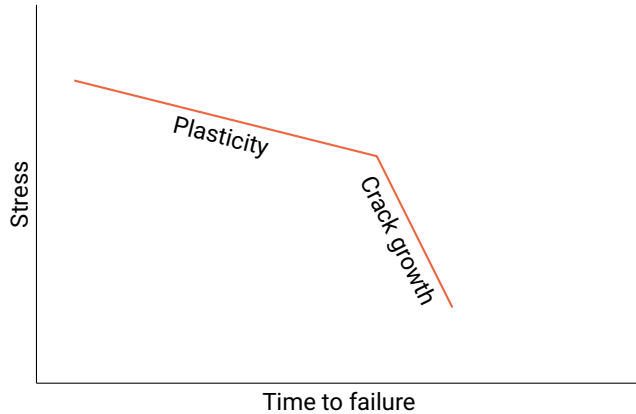
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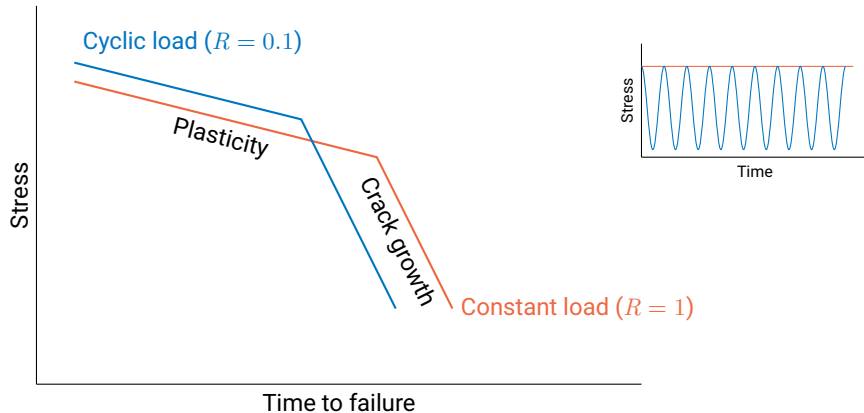
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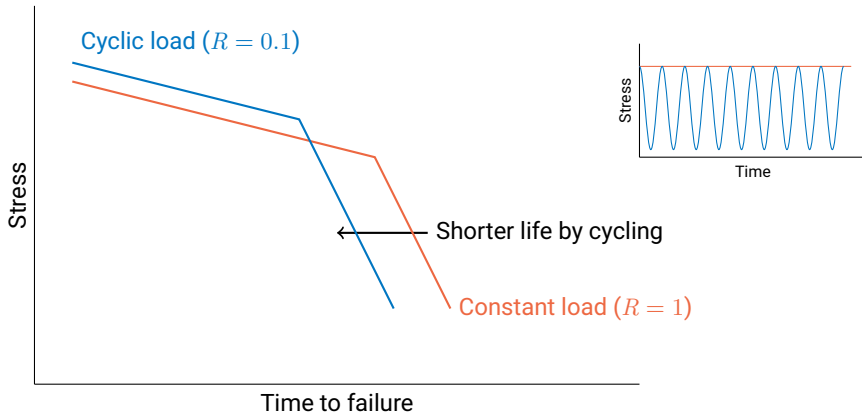
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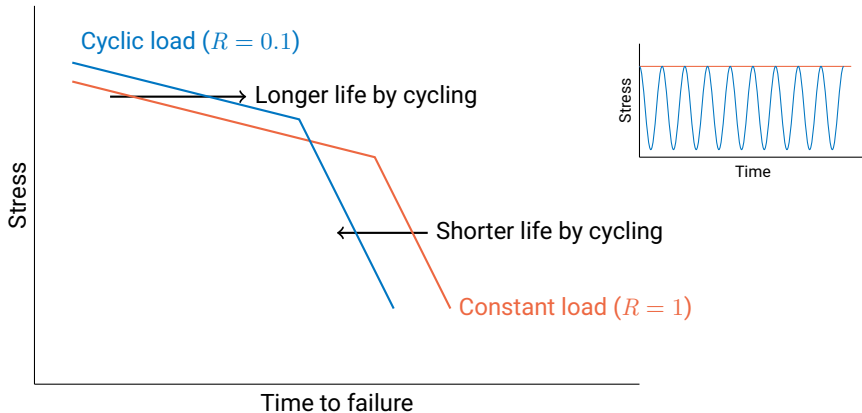
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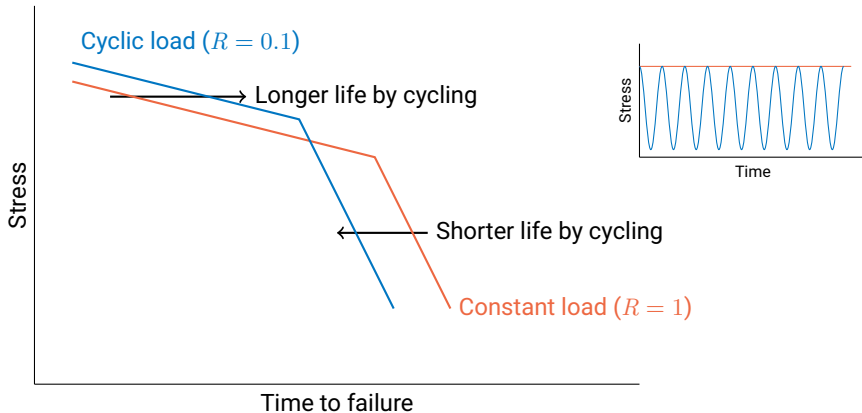
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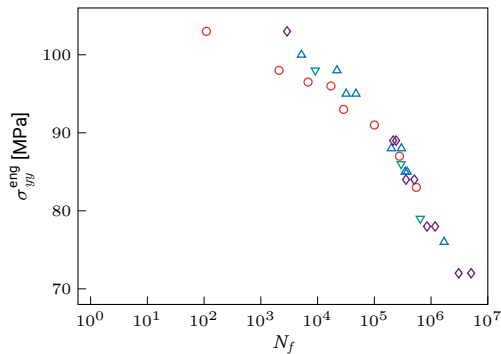
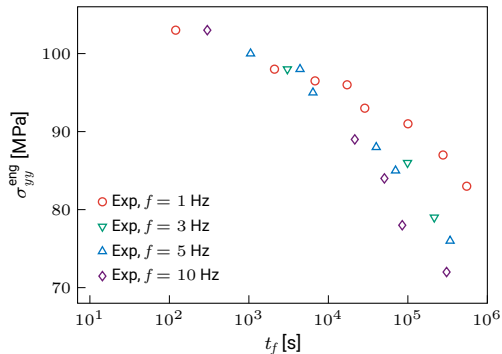
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Are these phenomena also observable in composites?

Frequency dependence confirms the transition between the different mechanisms

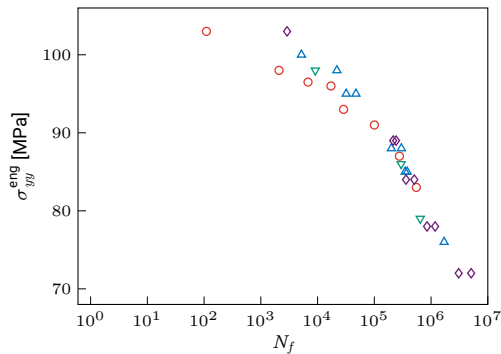
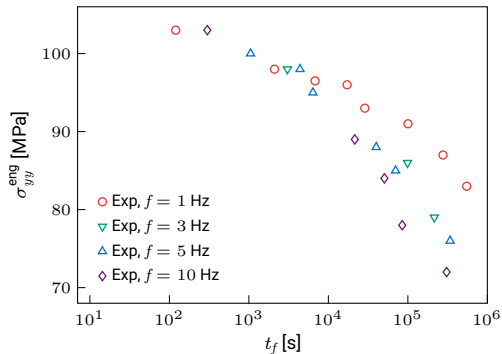
Testing 90° carbon/PEEK until failure, at $R = 0.1$ and different frequencies



- Plasticity controlled: time to failure is frequency-independent
- Crack growth controlled: number of cycles to failure is frequency-independent

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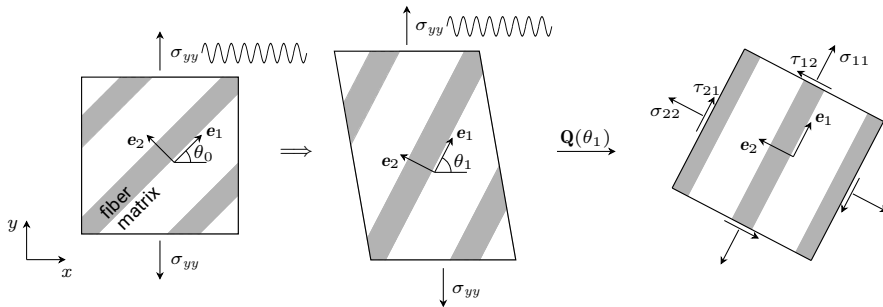
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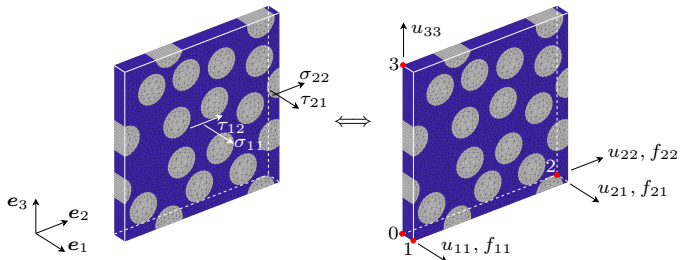
How can these phenomena be modeled?

Micromechanical modeling of fatigue in thermoplastics



Aim: model cyclic off-axis response

- Thin slice RVE
- BCs for off-axis loading
- Viscous model for the matrix
- Interelement cracks
- Fatigue cohesive zone model



Eindhoven Glassy Polymer (EGP) material model for the polymer matrix¹

Features of the model:

- Additive split of the Cauchy stress

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_h + \boldsymbol{\sigma}_r + \boldsymbol{\sigma}_d$$

- Hydrostatic part is hyperelastic

$$\boldsymbol{\sigma}_h = \kappa (\det(\mathbf{F}) - 1) \mathbf{I}$$

- Hardening part represents polymer chain reorientation

$$\boldsymbol{\sigma}_r = G_r \bar{\mathbf{B}}$$

- Driving stress is the sum from all Maxwell elements

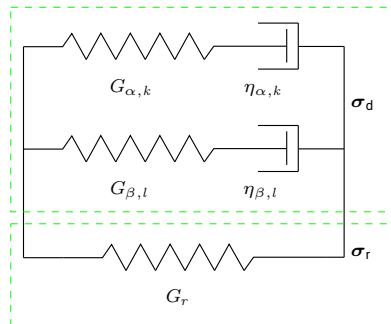
$$\boldsymbol{\sigma}_d = \sum_{\xi} \sum_j G_{\xi,j} \bar{\mathbf{B}}_e^{\xi,j}$$

- Multiplicative decomposition of $\mathbf{F}$

$$\mathbf{F} = \mathbf{F}_e^{\xi,j} \cdot \mathbf{F}_p^{\xi,j}$$

- Non-Newtonian flow for plastic deformations

$$\mathbf{D}_p^{\xi,j} = \frac{\boldsymbol{\sigma}_d^{\xi,j}}{\eta(\tau_{\xi}, p, S_{\xi}, T)}$$



¹Van Breemen et al., *J Mech Phys Solids*, 2011

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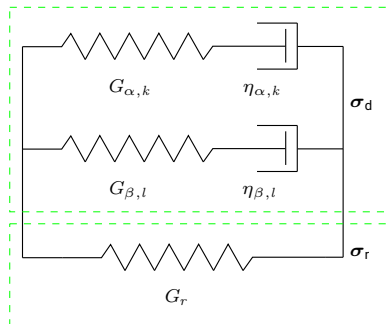
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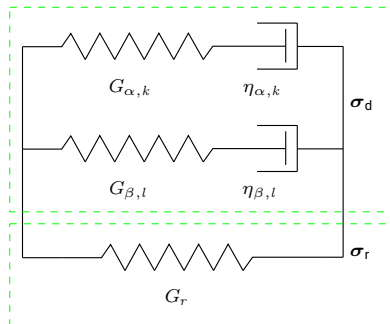
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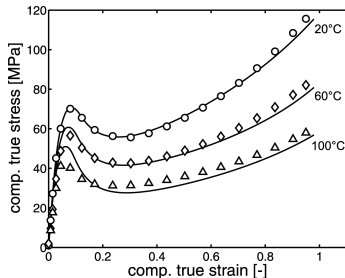
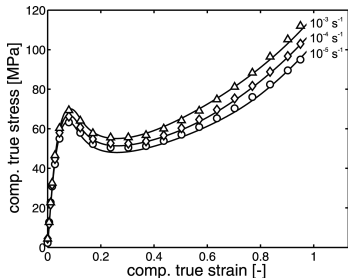
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The EGP model describes rate and temperature dependent behavior for large strains

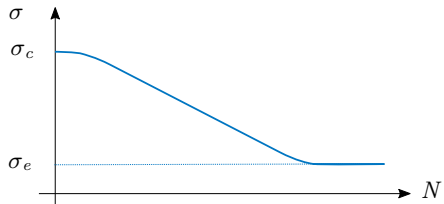
Example: uniaxial compression of polycarbonate¹



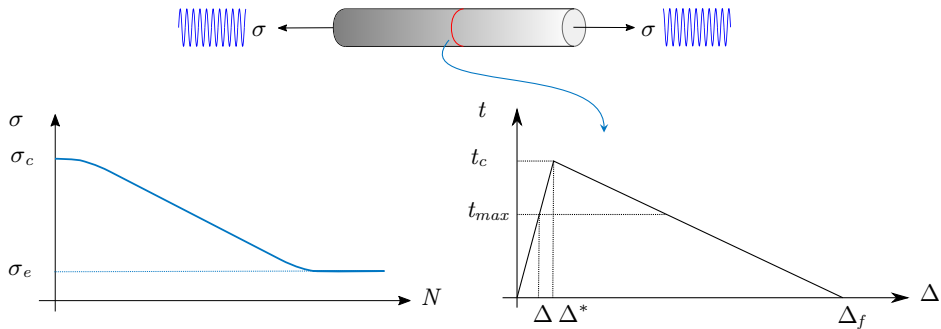
Number of α modes: 17; number of β modes: 1

¹Senden et al., *J Polym Sci B: Polym Phys*, 2012

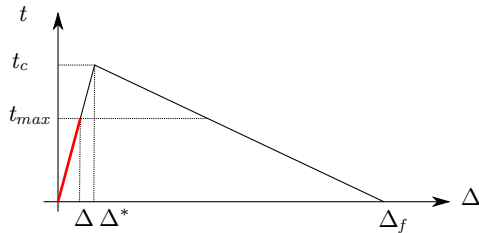
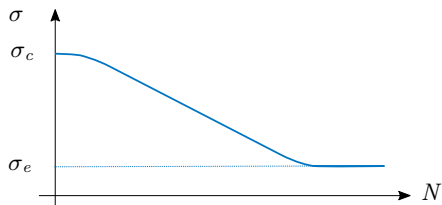
Fatigue cohesive zone model that covers initiation and propagation



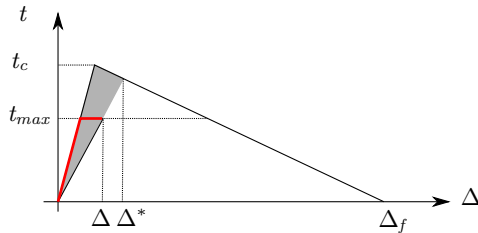
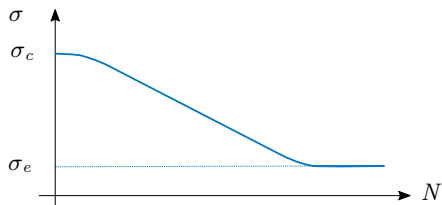
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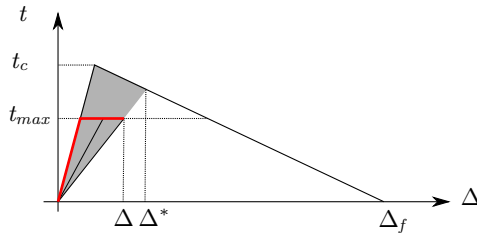
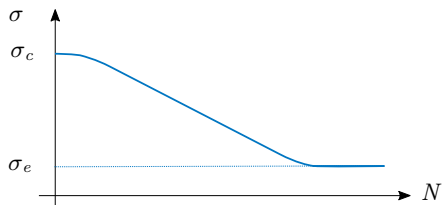
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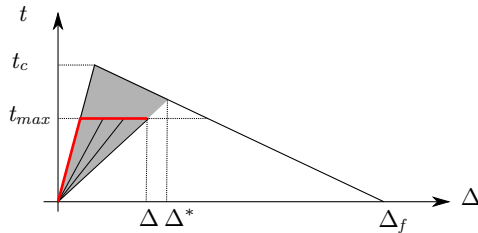
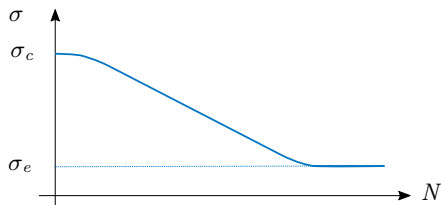
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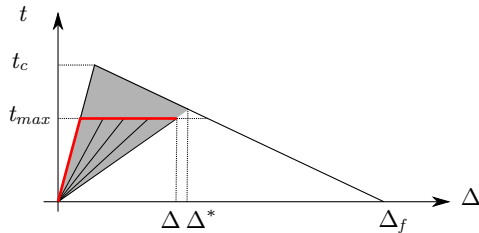
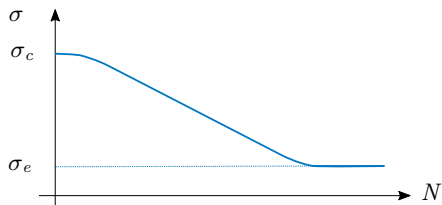
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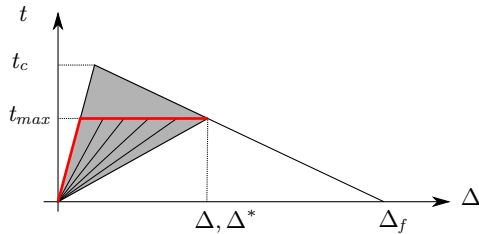
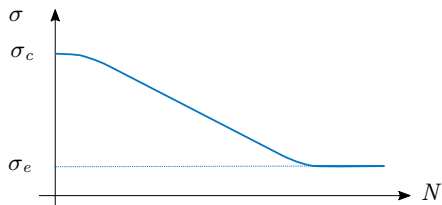
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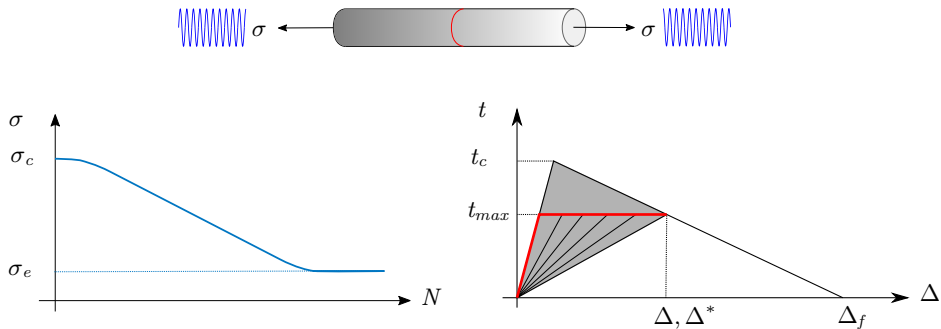
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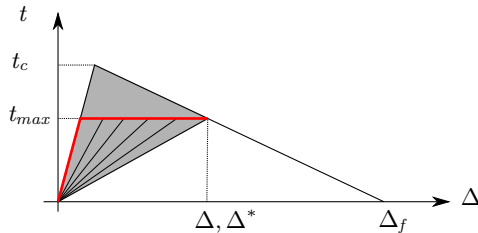
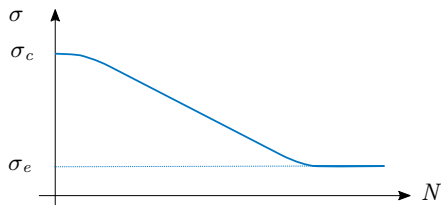


Fatigue cohesive zone model that covers initiation and propagation



S-N data as input, Paris curve as output

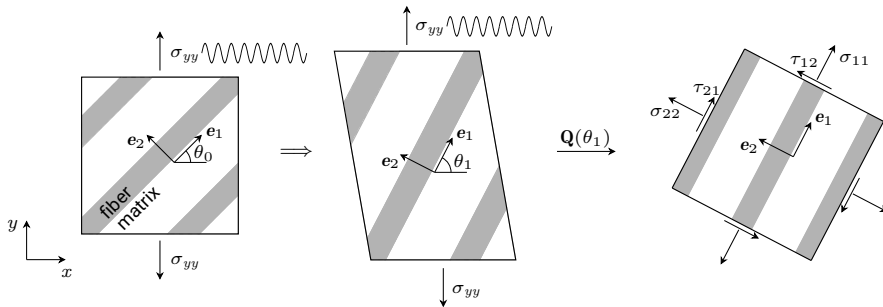
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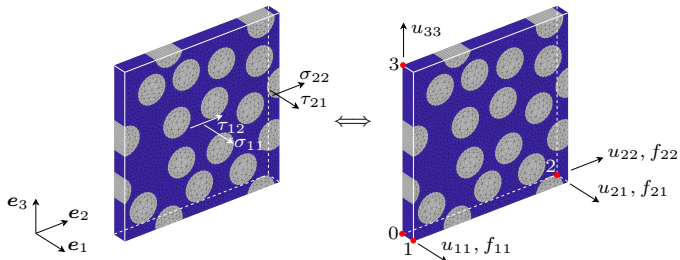
Dependent on mode of fracture and stress ratio (R)

Micromechanical modeling of fatigue in thermoplastics

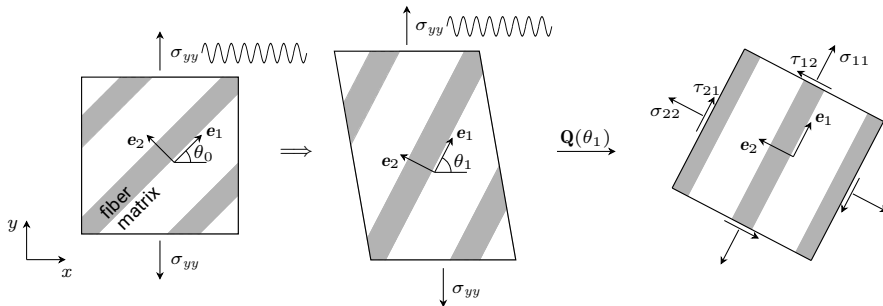


Aim: model cyclic off-axis response

- Thin slice RVE
- BCs for off-axis loading
- Viscous model for the matrix
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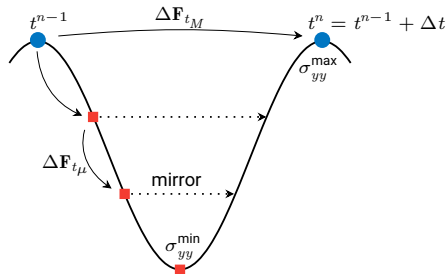


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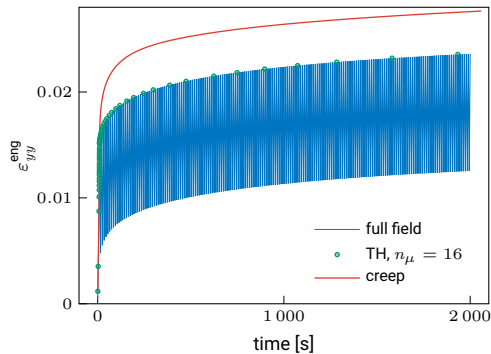
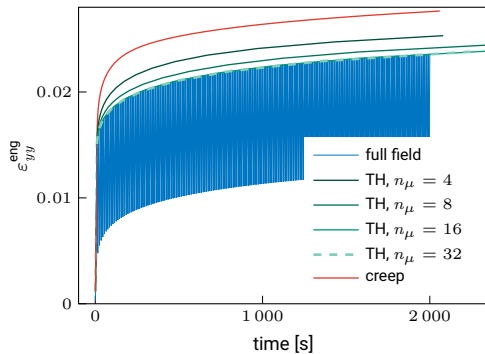
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Time-homogenization effectively speeds up the simulation

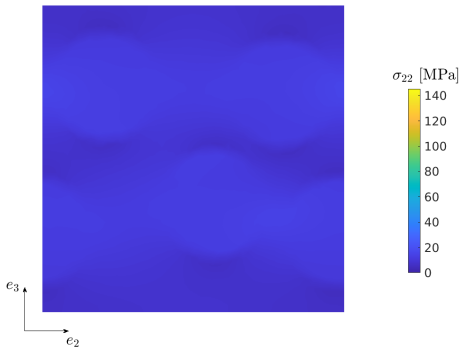
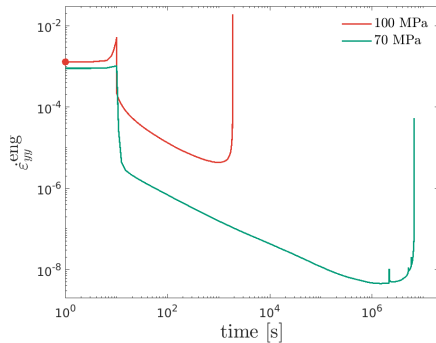
Comparing fully-resolved simulations to those with time-homogenization (TH)



- Sufficient micro time steps need to be taken for each resolved cycle (n_μ)
- Macro time steps are adaptively determined and can be much larger than the cycle period

Failure is detected as a quick increase in strain-rate

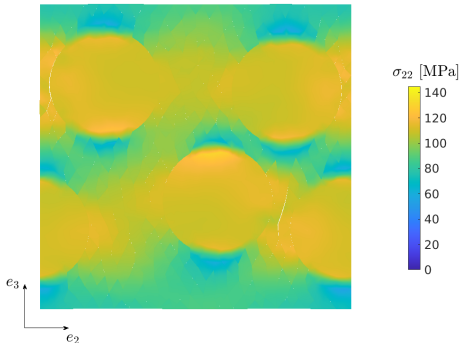
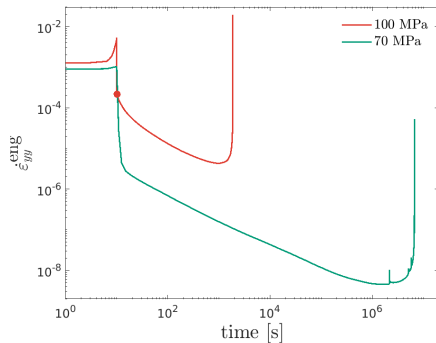
$$\chi = 90^\circ, \sigma_{yy}^{\text{eng}} = 100 \text{ MPa}, R = 0.1, f = 1 \text{ Hz}$$



The simulation terminates when the material cannot carry the applied load

Failure is detected as a quick increase in strain-rate

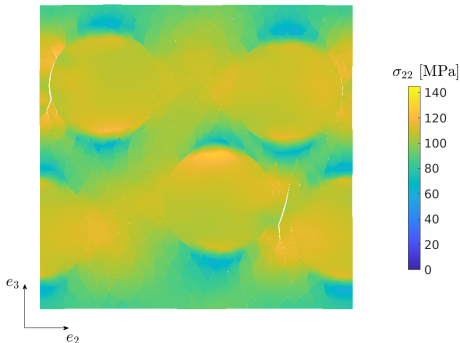
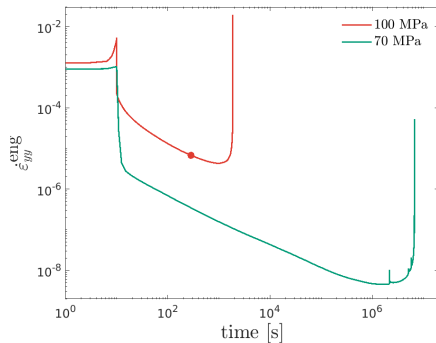
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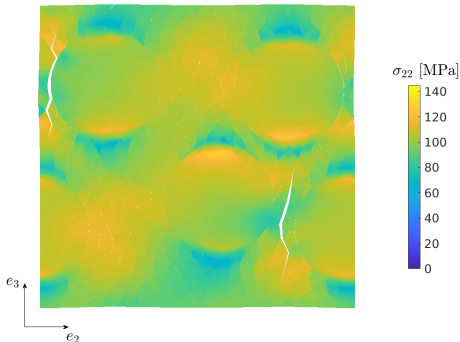
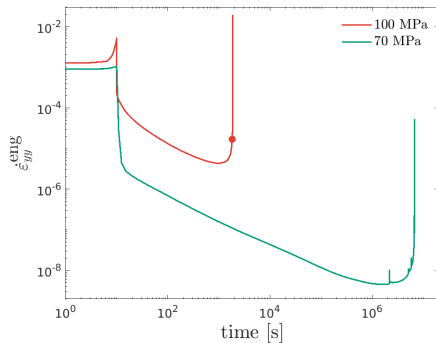
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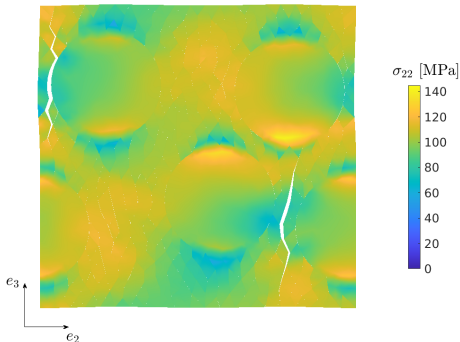
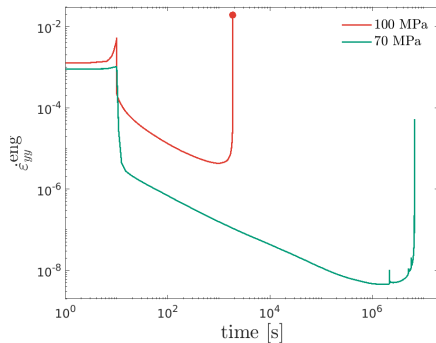
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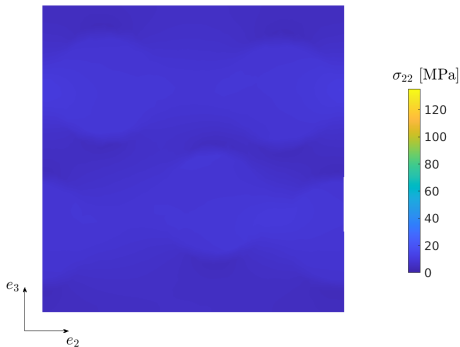
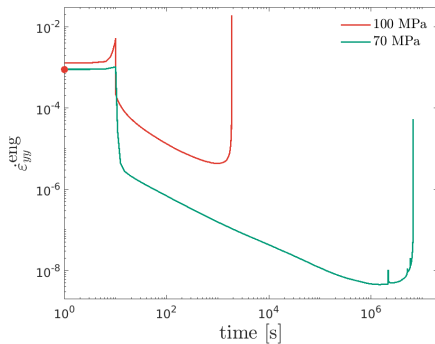
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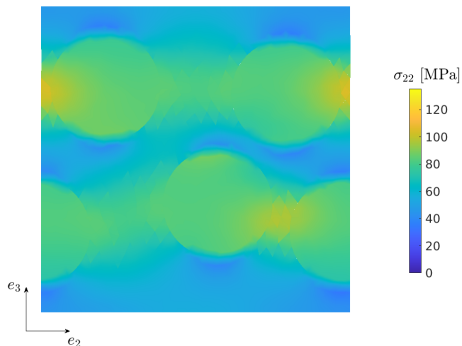
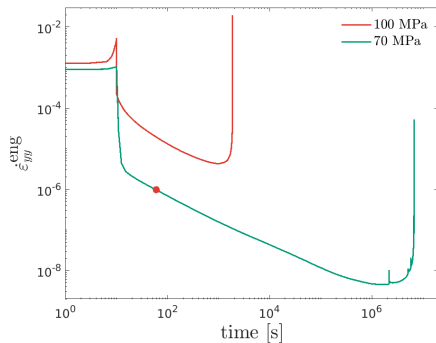
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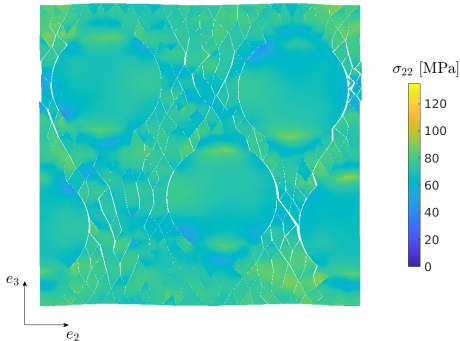
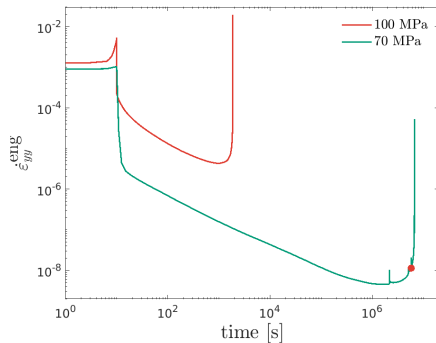
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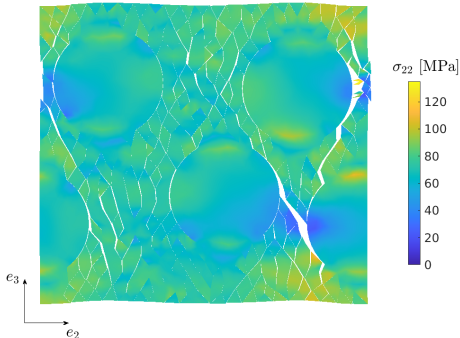
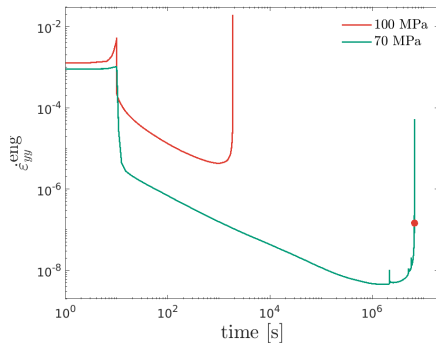
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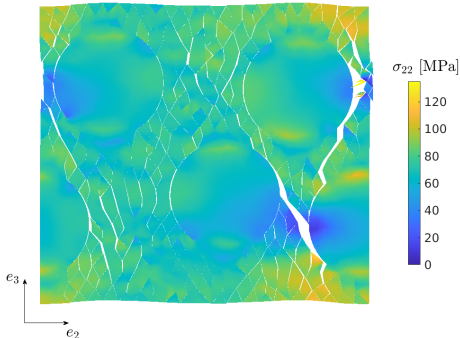
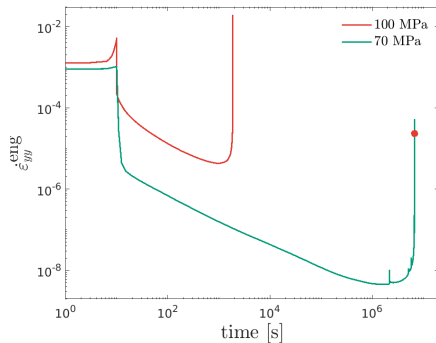
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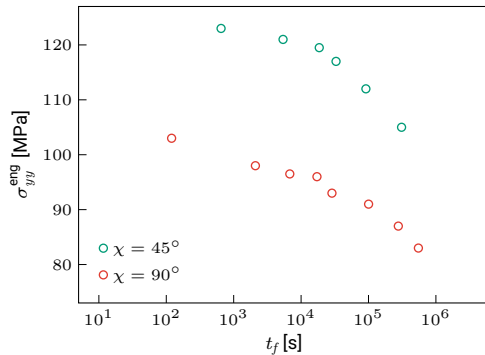
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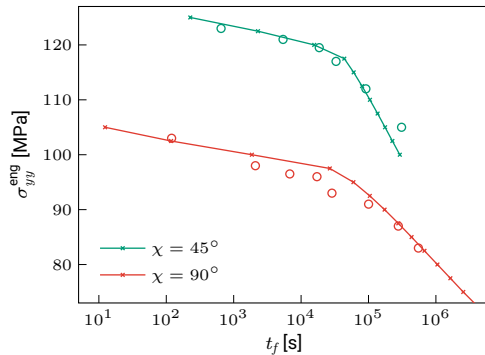
The model is calibrated on a single frequency with two different angles

Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



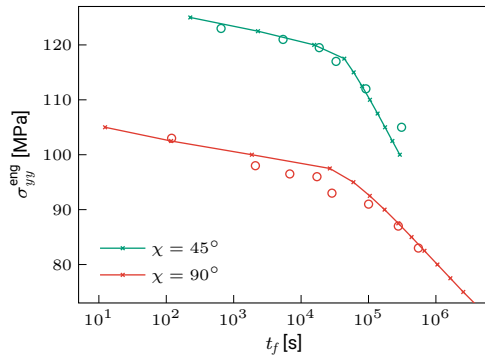
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Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



The model is calibrated on a single frequency with two different angles

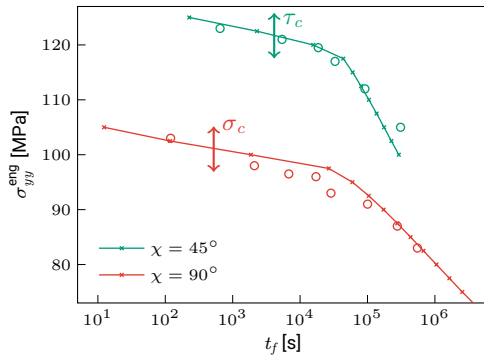
Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



- EGP spectrum from earlier creep simulations

The model is calibrated on a single frequency with two different angles

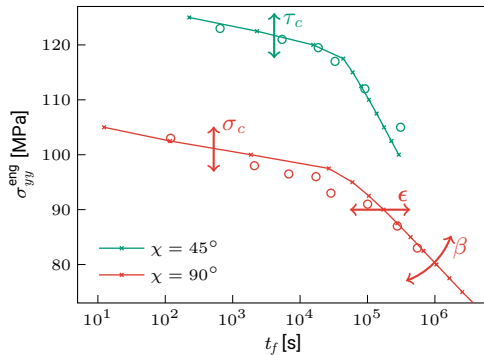
Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



- EGP spectrum from earlier creep simulations
- Cohesive strength to match plasticity regime
 - Tensile strength σ_c for 90°
 - Shear strength τ_c for 45°

The model is calibrated on a single frequency with two different angles

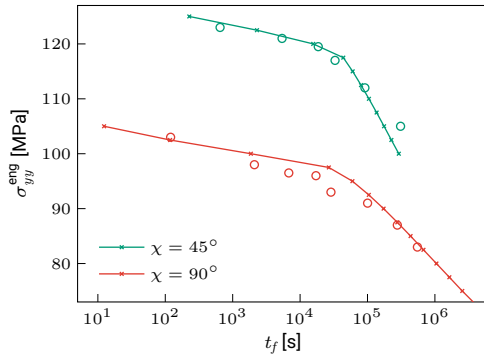
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- Fatigue parameters for crack growth regime
 - Endurance limit ϵ for offset
 - S-N parameter β for slope

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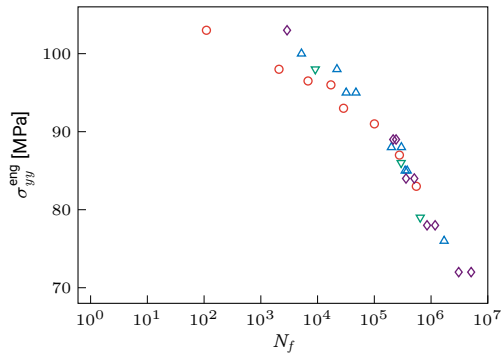
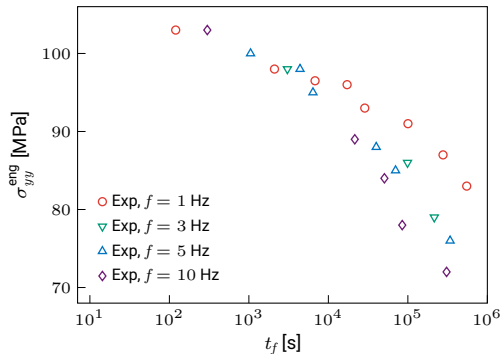


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- Transition between regimes can be improved with recalibration of EGP spectrum
- Dependence on fracture mode in the cohesive model needs more attention

Transition from plasticity controlled to crack growth controlled failure is captured

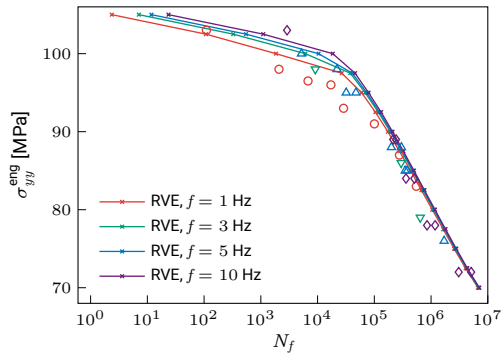
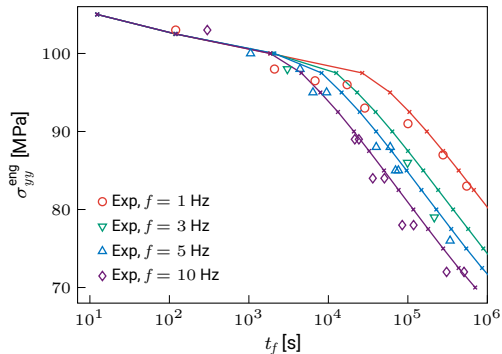
Simulation of 90° tests at $R = 0.1$ and different frequencies



- Plasticity controlled: time to failure is frequency-independent
- Crack growth controlled: number of cycles to failure is frequency-independent

Transition from plasticity controlled to crack growth controlled failure is captured

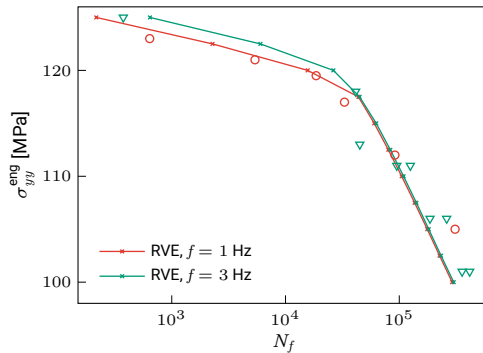
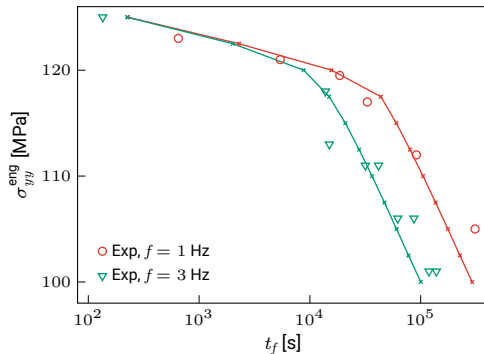
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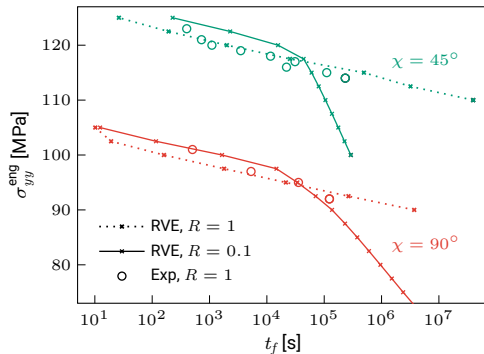
Simulation of 45° tests at $R = 0.1$ and different frequencies



- Plasticity controlled: time to failure is frequency-independent
- Crack growth controlled: number of cycles to failure is frequency-independent

Creep rupture is also captured

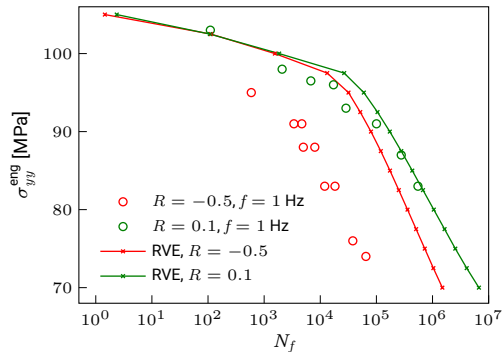
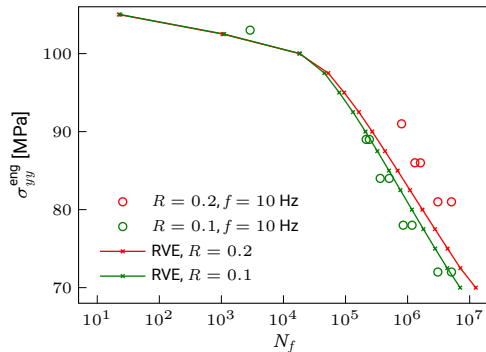
Testing samples at $R = 1$, simulations without fatigue damage and time-homogenization



- Shorter lifetime for increasing R in plasticity controlled failure
- Elastic opening of cohesive cracks plays a role in the response

R -dependence in crack-growth regime is not so well captured

Testing 90° samples at $R = 0.2$ and $R = -0.5$



- Longer lifetime for increasing R in crack growth controlled failure
- R -dependence is underpredicted by the model and comes with a change in slope

Conclusions & discussion

A unified micromechanical framework for modeling creep and fatigue in composites

- ✓ Large deformation nonlinear visco-elastic polymer model: EGP
- ✓ Cohesive zone model with cyclic damage: Dávila
- ✓ Time-homogenization for efficient time stepping with cyclic load

The model reproduces experimentally observed phenomena from off-axis tests

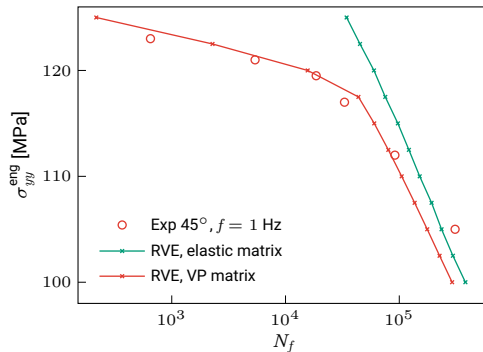
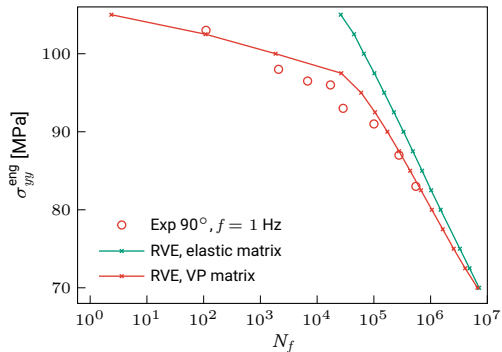
- Two failure regimes with different frequency dependence are distinguished in fatigue simulations
- Creep rupture ($R = 1$) is captured with the same model
- Improvement is possible in dependence on R -value and mode of fracture

[Kovačević et al (2024), *JMPS*]



Without plasticity no transition is obtained

Simulations are repeated with a hyperelastic material for the matrix



- The only remaining mechanism is crack-growth controlled failure
- Also in the crack-growth controlled domain there is influence of plasticity